REVERSE

GRAY HAIR

Scalp Serum

One recommended development formula / proposed 50 mL product



Selected visual: the gray-to-blue VETUD concept, updated to PEPTIDE SERIES with Darkenyl, Arcolys, SILVERFREE and pro-vitamin B5 highlights. Proposed artwork; the formula on page 2 is the technical specification.

DARKENYL 1% + ARCOLYS 1% + SILVERFREE 1.5%

Commercial blends as supplied. New combination remains unvalidated.

Recommended serum formula

Code VTD-RGH-S01 / one product proposal. Percentages and pilot quantities are calculated by weight.

Ingredient	% w/w	1 kg pilot / g	Purpose
Darkenyl commercial blend	1.00	10.00	Pigmentation candidate [2] [5]
Arcolys commercial blend	1.00	10.00	Botanical candidate [4]
SILVERFREE commercial blend	1.50	15.00	Peptide candidate [3]
Propanediol	4.00	40.00	Humectant
Glycerin	2.00	20.00	Humectant
Betaine	1.00	10.00	Humectant
Panthenol, active-content basis	0.50	5.00	Scalp conditioning
Hydroxyethylcellulose	0.25	2.50	Texture
Allantoin	0.10	1.00	Conditioning
Phenoxyethanol / ethylhexylglycerin 90:10 preservative blend	1.00	10.00	Proposed preservation
Purified water, before pH adjustment	87.65	876.50	Final water q.s.
Nominal total	**100.00**	**1000.00**	Before pH adjustment

Final balance: use suitable pH-adjustment solution as needed and subtract its mass from the water allowance. Provisional pH target is 5.0-5.5, conditional on the current technical sheets for all three actives.

50 mL bottle: measure finished-product density. Fill mass (g) = 50 mL x density (g/mL). Ingredient mass per bottle = fill mass x ingredient percentage / 100. Do not equate 50 mL with 50 g without measurement.

Meaning of the percentages: 1.5% SILVERFREE is 1.5% supplied blend, not 1.5% pure Palmitoyl Dipeptide-52. Account for all water, solvents and preservatives already carried by the blends.

Confirm the active ingredients

This is the recommended candidate to assess, not an established superior formula. Lock the trade grades before compounding.

Darkenyl / proposed 1.00% w/w

Obtain the current Givaudan INCI, active specification, density, preservative content and technical sheet. The published associated lotion used 1% v/v. Matching that concentration by weight requires both blend and finished-product density; confirm the proposed 1% w/w target before the pilot. Verify that the supplied grade corresponds to the study material. [2] [5]

Arcolys / proposed 1.00% supplied blend

Use the traceable commercial material, not an unspecified root extract. Ask Greentech for its complete clinical dossier and current standardization, storage, pH and temperature limits. The available clinical report supports investigating this ingredient; it does not validate the new three-active combination. [4]

SILVERFREE / proposed 1.50% supplied blend

Croda lists Glycerin, Water and Palmitoyl Dipeptide-52 and recommends 1.5% use. Request the full human-study reports, exact supplied-grade specification and stability requirements. Confirm peptide-content disclosure where available. Avoid inferring pure-peptide concentration from the blend addition. [3]

Vehicle and preservation

Select a hydroxyethylcellulose grade for a low-residue scalp serum. Adjust panthenol powder or liquid weights to deliver 0.50% active. Calculate cumulative preservatives from all blends. Confirm compatibility across the final pH range and test the actual bottle, closure and applicator.

Build and verify the pilot

The chemist should finalize the processing order and limits from the chosen raw-material grades.

1 / Prepare the base

Weigh the water reserve and pre-disperse/hydrate the polymer using the selected grade's recommended procedure. Dissolve the humectants and conditioning ingredients with an appropriate validated method. Resolve any allantoin crystals or polymer agglomerates before adding the branded actives.

2 / Incorporate the actives

Add the supplied blends one at a time within the lowest permitted temperature limit in their technical sheets. Record appearance, pH, clarity and viscosity after each addition. Incorporate the selected preservative system at its recommended stage. Do not adopt a heating process until active-grade limits are confirmed.

3 / Set the final bulk

Adjust pH only after checking the shared compatibility window. Bring to final batch weight with purified water, mix uniformly and deaerate if necessary. Record density, appearance, odor, pH, viscosity and batch yield. Set provisional specifications using actual pilot measurements, not arbitrary acceptance values.

4 / Establish release and shelf-life evidence

Perform preservation challenge, microbial quality and packaging-compatibility work. Assess stability under relevant storage and transport conditions, including pH drift, precipitation, discoloration and active retention where methods are available. Confirm scalp tolerability under intended use. Shelf life and instructions remain to be established.

Study handoff: have a qualified research team assess the finished formula against a matched vehicle over about four to six months. Predefine gray-hair density in consistent scalp areas, standardized photographs, staining controls, tolerability, sample-size justification and the analysis of dropouts.

What supports this proposal

Ingredient evidence informs the candidate. It does not establish the finished VETUD serum's efficacy.

Controlled ingredient research

The Darkenyl-associated study included 44 men over four months. Mean gray-hair density decreased 14% with the active lotion versus 5% with placebo. The research was supplier-associated. Its methods and conclusion differ on use frequency; obtain clarification before matching the protocol. [2]

Supplier clinical summaries

The Arcolys report describes 44 men and women, 1% blend, and a 19% decrease in white-hair density from baseline after five months. That figure is not a placebo-adjusted benefit. Croda reports four SILVERFREE clinical studies, but all four full reports have not been independently reviewed here. [4] [3]

A finished-system benchmark

A 2026 Arey trial used four groups and 173 adults with 5-25% gray hair over 24 weeks. Its combination improved a scaled score by 13%; reported raw counts changed from 35.11 to 34.38. The serum-only arm showed little objective change. Brand funding, one-tailed tests and completer analysis limit interpretation. This was a different finished system. [1]

Claims to finalize after testing

No head-to-head trial establishes that this VETUD candidate is superior, and synergy of its three blends has not been demonstrated. Treat "Reverse Gray Hair" as a working name. Claims that alter body structure or function can change US classification; review the final name, label and advertising for every launch market. [6]

What the chemist should return

One product formula, one confirmed raw-material set, and a documented development decision.

Requested deliverables

Return the revised master formula in % w/w, exact trade grades and INCI breakdown, manufacturing method with justified limits, pilot observations, measured fill density, provisional specifications, preservation strategy, test plan, stability plan and a costed 50 mL packaging proposal. List unresolved compatibility or efficacy questions explicitly.

[1] Bhardwaj, V., et al. (2026).
Randomized double-blind, placebo-controlled study of a supplement and serum on grey hair. Journal of Drugs in Dermatology.
Journal paper / DOI

[2] De Tollenaere, M., et al. (2021).
Global repigmentation strategy of grey hair follicles by targeting oxidative stress and stem cells protection. Applied Sciences, 11, 1533.
Journal paper / DOI

[3] Croda Beauty / Sederma.
SILVERFREE product information and supplier use-level guidance.
Official ingredient page

[4] Schlegel, P. / Greentech (2021).
Reverse hair greying. Supplier-hosted Arcolys clinical summary, COSSMA.
Clinical summary PDF

[5] Givaudan Active Beauty.
Darkenyl product information; current technical dossier needed for grade matching.
Official ingredient page

[6] US Food and Drug Administration.
Is it a cosmetic, a drug, or both? Intended-use and claims framework.
FDA explanation

Document status: development proposal prepared for VETUD, 09 November 2026. Formula safety, preservation, shelf life and gray-hair efficacy remain unestablished. The single supplied picture is creative reference, not an approved label or proof of outcome.